

RH-GHM-NCHG

Soft Proof v0.97

Arithmetic Vertical Connection, Derived Hardy Completion, and the Single Remaining Graph-Closure Obligation

Epistemic status. This document freezes the research state reached after replacing the Naive Proof's single Global Relative Verticality (GRV) assumption by an explicit arithmetic connection identity. The differential/operator content of GRV is derived here on a zero-blind Mellin core. The remaining gap is isolated to one global derived Hardy-completion statement, labelled **AVC-2**. This is therefore a *soft-proof candidate checkpoint*, not an unconditional proof of the Riemann Hypothesis.

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Companion to the RH-GHM-NCHG Research Ledger, Supplemental I, C2 Ledger, and Naive Conditional Proof.

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1 Purpose and status freeze

The Naive Conditional Proof reduced the analytic endgame to a single zero-blind structural hypothesis, Global Relative Verticality (GRV): the NCHG vertical Cartan generator should realize as the Mellin normal derivative preserving Burnol's arithmetic Hardy subspace. In that document the exact local/history identities were already closed, while GRV remained one global black box.

The present checkpoint opens that box. Its main new result is an explicit arithmetic connection

$$\nabla_{\text{ar}} = \mathbf{L} + \mathbf{R} - \mathbf{P}_{\Lambda} \quad (1)$$

for which the normalized Poisson/Mellin map satisfies

$$\partial_s \mathcal{E} = \mathcal{E} \nabla_{\text{ar}} \quad (2)$$

on a zero-blind Mellin core. This is the constructive differential content formerly hidden inside GRV.

The residual global problem is no longer “there exists a physical map” or “zeros are physical”. It is the following graph-closure/no-loss question: can the exact NCHG/history completion be extended through the Burnol Hardy completion without creating a new normal component? This remaining statement is **AVC-2**.

Short status.

- Naive GRV black box: **REPLACED** by AVC-1 + AVC-2.
- AVC-1 (explicit arithmetic vertical connection): **PROVED HERE** on a natural zero-blind Mellin core.
- NCHG transverse exactness: **VERIFIED** from the C2 Ledger.
- Low-arity history/ray exact square: **VERIFIED**.
- Burnol Hardy factorization and $B = 1 \Leftrightarrow \text{RH endpoint}$: **EXTERNAL INPUT**, carried from the Naive Proof.
- AVC-2 (derived Hardy completion/no-loss): **OPEN BOSS**.
- AVC-1 + AVC-2 $\Rightarrow$ derivative invariance $\Rightarrow B = 1 \Rightarrow \text{RH}$: **PROVED CONDITIONALLY**.

2 Frozen inputs from the earlier ledgers

2.1 Seed algebra and derived transverse/phase plane

The nonassociative real algebra is

$$A = \text{span}_{\mathbb{R}}\{1, h, q\}, \quad h^2 = q, \quad q^2 = q, \quad hq = 1, \quad qh = -1. \quad (3)$$

On the two-dimensional quotient $V = A/\mathbb{R}1$, the verified split-Clifford triple is

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad (4)$$

with

$$H^2 = S^2 = I, \quad J^2 = -I, \quad HS = J = -SH. \quad (5)$$

The positive metric is

$$e(u, v) = -\omega(u, Jv). \quad (6)$$

The associator map $V \otimes V \rightarrow \text{End}(V)$ was independently verified to have determinant 3 in the natural basis, hence is an isomorphism over characteristic zero.

2.2 History complex and BRST backbone

The NCHG history complex is

$$\mathbb{H}_n = C_\bullet(K_n) \otimes A^{\otimes n}, \quad \mathbb{H} = \bigoplus_{n \geq 1} \mathbb{H}_n, \quad (7)$$

with cellular differential

$$d^2 = 0. \quad (8)$$

At arity three, the oriented reassociation edge α obeys

$$\boxed{d\alpha(x, y, z) = (xy)z - x(yz) = [x, y, z].} \quad (9)$$

The framework also contains homotopy-transfer data and a formal BRST physical quotient

$$\mathcal{H}_{\text{phys}} = \frac{\ker Q_{\text{BRST}}}{\text{im } Q_{\text{BRST}}}. \quad (10)$$

Full cyclic nondegeneracy and quantum anomaly cancellation remain separate NCHG obligations and are not used as hidden premises in AVC-1.

2.3 Normalized ray complex and exact transverse direction

Choose an oriented unit ray $r \in V$ and set $n = -Jr$. Define the graded 1-2-1 complex

$$C_r^{-1} = \mathbb{C}e_-, \quad C_r^0 = \mathbb{C}r \oplus \mathbb{C}n, \quad C_r^1 = \mathbb{C}e_+, \quad (11)$$

with

$$q_r e_- = r, \quad q_r r = 0, \quad q_r n = e_+, \quad q_r e_+ = 0. \quad (12)$$

In the ordered basis (e_-, r, n, e_+) ,

$$q_r = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad q_r^2 = 0. \quad (13)$$

The verified homotopies are

$$h_r = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad b_r = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad (14)$$

for which

$$q_r h_r + h_r q_r = I_4, \quad (15)$$

and

$$q_r b_r + b_r q_r = \text{diag}(1, 1, -1, -1). \quad (16)$$

Restricting the latter to $V = \text{span}\{r, n\}$ gives H .

With centered spectral parameter

$$s = \frac{1}{2} + \delta + i\gamma, \quad (17)$$

and $Q_\delta = \delta q_r$, one obtains

$$\boxed{\{Q_\delta, h_r\} = \delta I,} \quad \boxed{\{Q_\delta, b_r\}|_V = \delta H.} \quad (18)$$

The Hodge Laplacian is

$$\Delta_\delta = Q_\delta Q_\delta^\dagger + Q_\delta^\dagger Q_\delta = \delta^2 I. \quad (19)$$

Thus the transverse real parameter is not merely damped; it has an explicit Q -primitive.

2.4 The same Cartan operator is a history boundary

On V , the reconstructed associator operators are

$$A_{hh} = \begin{pmatrix} 0 & -1 \\ 0 & 1 \end{pmatrix}, \quad A_{hq} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \quad (20)$$

$$A_{qh} = \begin{pmatrix} -1 & 0 \\ -1 & -2 \end{pmatrix}, \quad A_{qq} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}. \quad (21)$$

They satisfy

$$A_{hq} + A_{qh} + A_{qq} = H. \quad (22)$$

Define the operator-valued history edge

$$\mathbf{b}_H(w) = \alpha(h, q, w) + \alpha(q, h, w) + \alpha(q, q, w). \quad (23)$$

Using (9),

$$\boxed{d_{\text{hist}} \mathbf{b}_H = H.} \quad (24)$$

Hence the transverse generator admits two explicit resolutions:

$$\boxed{H = d_{\text{hist}} \mathbf{b}_H = \{q_r, b_r\}|_V.} \quad (25)$$

3 Analytic endpoint inherited from the Naive Proof

The Naive Proof uses Burnol's Hardy-space closure/causality formulation as an external analytic endpoint. We keep that input explicit rather than re-proving it here.

Let

$$H_{1/2}^2 := H^2\left(\Re s > \frac{1}{2}\right), \quad (26)$$

and let E denote the relevant Poisson/adelic half-density map. Let

$$V(s) = \frac{s-1}{s} \quad (27)$$

be the fixed pole-normalizing factor, and define the normalized map

$$\mathcal{E} := VE. \quad (28)$$

We use the *closed* Burnol subspace

$$\boxed{\Delta := \overline{\mathcal{E}(\mathcal{S}_{\leq 1})}^{H_{1/2}^2}.} \quad (29)$$

The factorization input is

$$\boxed{\Delta = BH_{1/2}^2,} \quad (30)$$

where B is the relevant inner factor. The Naive Proof carries the published endpoint

$$\boxed{B = 1 \iff \text{RH}.} \quad (31)$$

For the soft-proof architecture, the only role of this external theorem is the final implication after derivative invariance has been established.

Closure convention. Earlier project documents sometimes wrote $\Delta = \mathcal{E}(\mathcal{S}_{\leq 1})$ schematically. In v0.97 we explicitly use the closed invariant subspace (29), because the remaining difficulty is precisely

a graph-closure/completion problem.

4 From GRV to an explicit arithmetic vertical connection

4.1 Mellin convention and zero-blind core

For φ in a compactly supported smooth Mellin core

$$\mathcal{D}_0 := C_c^\infty(0, \infty), \quad (32)$$

define

$$\widehat{\varphi}(s) = \mathcal{M}\varphi(s) := \int_0^\infty \varphi(x) x^s \frac{dx}{x}. \quad (33)$$

The normalized arithmetic map has Mellin symbol

$$\boxed{\mathcal{M}(\mathcal{E}\varphi)(s) = V(s)\zeta(s)\widehat{\varphi}(s),} \quad (34)$$

initially on the Euler half-plane. This identity is an analytic input inherited from the normalized Poisson/Mellin construction used in the Naive Proof.

The core $\mathcal{D}_0$ is chosen only to make every operator calculation below absolutely legitimate. The difficult extension from this core to Burnol's source class $\mathcal{S}_{\leq 1}$ is deliberately isolated into AVC-2 rather than smuggled into AVC-1.

4.2 Three explicit source operators

Define the log-time tangent operator

$$(\mathbf{L}\varphi)(x) = (\log x)\varphi(x). \quad (35)$$

Define the pole/half-form correction

$$(\mathbf{R}\varphi)(x) = \frac{1}{x} \int_x^\infty \varphi(y) dy - \int_x^\infty \varphi(y) \frac{dy}{y}. \quad (36)$$

Define the prime-power operator

$$(\mathbf{P}_\Lambda\varphi)(x) = \sum_{n \geq 2} \Lambda(n)\varphi(nx), \quad (37)$$

where Λ is the von Mangoldt function. For compactly supported φ , the sum is pointwise finite for each $x > 0$.

4.3 Exact Mellin transforms

For $\mathbf{L}$,

$$\widehat{\mathbf{L}\varphi}(s) = \partial_s \widehat{\varphi}(s). \quad (38)$$

For the first integral in (36), Fubini gives for $\Re s > 1$:

$$\int_0^\infty x^{s-1} \left(\frac{1}{x} \int_x^\infty \varphi(y) dy \right) dx = \int_0^\infty \varphi(y) \left(\int_0^y x^{s-2} dx \right) dy \quad (39)$$

$$= \frac{1}{s-1} \widehat{\varphi}(s). \quad (40)$$

For the second integral,

$$\int_0^\infty x^{s-1} \left(\int_x^\infty \varphi(y) \frac{dy}{y} \right) dx = \int_0^\infty \frac{\varphi(y)}{y} \left(\int_0^y x^{s-1} dx \right) dy \quad (41)$$

$$= \frac{1}{s} \widehat{\varphi}(s). \quad (42)$$

Hence

$$\widehat{\mathbf{R}\varphi}(s) = \left(\frac{1}{s-1} - \frac{1}{s} \right) \widehat{\varphi}(s) = \frac{V'(s)}{V(s)} \widehat{\varphi}(s). \quad (43)$$

For the prime-power operator, a change of variables gives

$$\widehat{\mathbf{P}_\Lambda \varphi}(s) = \sum_{n \geq 2} \Lambda(n) \int_0^\infty \varphi(nx) x^s \frac{dx}{x} \quad (44)$$

$$= \sum_{n \geq 2} \frac{\Lambda(n)}{n^s} \widehat{\varphi}(s). \quad (45)$$

On $\Re s > 1$,

$$\sum_{n \geq 2} \frac{\Lambda(n)}{n^s} = -\frac{\zeta'(s)}{\zeta(s)}, \quad (46)$$

so

$$\widehat{\mathbf{P}_\Lambda \varphi}(s) = -\frac{\zeta'(s)}{\zeta(s)} \widehat{\varphi}(s). \quad (47)$$

No zero location enters the definition of $\mathbf{P}_\Lambda$; it is built directly from prime powers.

4.4 AVC-1 theorem

Define the arithmetic vertical connection by (1). Combining (38), (43), and (47),

$$\widehat{\nabla_{\text{ar}} \varphi}(s) = \left(\partial_s + \frac{V'(s)}{V(s)} + \frac{\zeta'(s)}{\zeta(s)} \right) \widehat{\varphi}(s). \quad (48)$$

Multiplying by $V(s)\zeta(s)$,

$$V\zeta \widehat{\nabla_{\text{ar}} \varphi} = V\zeta \partial_s \widehat{\varphi} + (V'\zeta + V\zeta') \widehat{\varphi} \quad (49)$$

$$= \partial_s (V\zeta \widehat{\varphi}). \quad (50)$$

Using (34) gives:

Theorem / Proposition

AVC-1 (Arithmetic Vertical Connection Identity). For every $\varphi \in \mathcal{D}_0 = C_c^\infty(0, \infty)$ and initially for $\Re s > 1$,

$$\partial_s \mathcal{M}(\mathcal{E}\varphi) = \mathcal{M}(\mathcal{E} \nabla_{\text{ar}} \varphi). \quad (51)$$

Equivalently, on this zero-blind Mellin core,

$$\partial_s \mathcal{E} = \mathcal{E} \nabla_{\text{ar}}. \quad (52)$$

No Riemann zero ordinate is used in the construction.

4.5 What AVC-1 has actually removed from GRV

The Naive Proof's GRV assumption schematically demanded that the NCHG vertical direction be realized by the Mellin normal derivative. AVC-1 gives the arithmetic side of that statement constructively: ∂_s is not postulated; it is the image under $\mathcal{E}$ of the explicit source connection (1).

The remaining issue is therefore not the existence of a vertical arithmetic derivative. It is whether the exact/history completion required to interpret the transverse part as Q -exact survives the Hardy-space completion without generating a new normal component.

5 Structural decomposition of the arithmetic connection

Write

$$\boxed{\nabla_{\text{ar}} = L + C_{\text{tr}}, \quad C_{\text{tr}} := R - P_{\Lambda}.} \quad (53)$$

The two terms have different conceptual jobs:

- $L = (\log x) \cdot$ is the ordinary log-time tangent whose Mellin action is ∂_s .
- R carries the $s = 0, 1$ pole/half-form normalization through V'/V .
- $-P_{\Lambda}$ carries the prime-power contribution; after Mellin transform it becomes $+\zeta'/\zeta$ in the connection.

Thus the arithmetic transverse/conductor correction is explicitly

$$\boxed{(C_{\text{tr}}\varphi)(x) = \frac{1}{x} \int_x^\infty \varphi(y) dy - \int_x^\infty \varphi(y) \frac{dy}{y} - \sum_{n \geq 2} \Lambda(n) \varphi(nx).} \quad (54)$$

Interpretive compression. The previous GRV language “ H maps to ∂_s ” is too coarse if read as an ordinary representation equality. The type-correct reading is that the NCHG vertical Cartan direction supplies the *vertical connection class*, while ∂_s is the normal connection on the analytic family. AVC-1 constructs the latter from log-time, pole, and prime-power data.

6 Exact lift of the transverse arithmetic correction

The C2 Ledger already proves that the normalized ray complex contracts the identity. Therefore any auxiliary operator A can be lifted into an exact ray differential.

Let A be a densely defined operator on an auxiliary carrier $\mathcal{K}$, and define

$$Q_A := q_r \widehat{\otimes} A. \quad (55)$$

Because $q_r^2 = 0$,

$$Q_A^2 = 0. \quad (56)$$

Using (15),

$$\boxed{\{Q_A, h_r \otimes I\} = I \otimes A.} \quad (57)$$

Using (16), on the middle ray plane,

$$\boxed{\{Q_A, b_r \otimes I\}|_V = H \otimes A.} \quad (58)$$

Apply this formally to the explicit arithmetic correction $A = C_{\text{tr}}$ on a common invariant core:

$$Q_{\text{tr}} := q_r \widehat{\otimes} C_{\text{tr}}. \quad (59)$$

Then

$$I \otimes \mathbf{C}_{\text{tr}} = \{Q_{\text{tr}}, h_r \otimes I\}. \quad (60)$$

Thus the *operator type* required for transverse exactness exists explicitly. What is not yet proved is that this formal exact lift is compatible with Burnol's global Hardy completion and its closed subspace Δ .

6.1 Compatibility with the history boundary

Equation (24) shows that the same Cartan direction is already a history boundary. The soft-proof target is therefore not to invent a new ghost mechanism, but to extend the following low-arity pattern through the arithmetic completion:

$$\text{history primitive} \mapsto \text{arithmetic primitive}, \quad H = d_{\text{hist}} \mathbf{b}_H \mapsto \mathbf{C}_{\text{tr}} = \{Q_{\text{tr}}, h_r \otimes I\}. \quad (61)$$

The unresolved question is topological/analytic: whether this exactness remains invisible in the normal quotient after completion.

7 The remaining open theorem: AVC-2

7.1 Why a completion theorem is necessary

AVC-1 is proved on $\mathcal{D}_0 = C_c^\infty(0, \infty)$ in the Euler domain. Burnol's analytic endpoint, however, lives in a closed Hardy subspace $\Delta = BH_{1/2}^2$ obtained from the source class $\mathcal{S}_{\leq 1}$ after Poisson/Mellin transformation and completion. Passing from a core identity to invariance of a closed subspace requires domain, graph-norm, and no-loss control.

The remaining theorem must therefore answer two questions at once:

1. Does ∇_{ar} preserve the Burnol source class modulo the NCHG/GHM exact sector?
2. Does adding those exact source corrections leave the closed Hardy image unchanged?

7.2 Condensed AVC-2 formulation

Let $\widetilde{\mathcal{S}}_{\leq 1}$ be a derived source completion containing $\mathcal{S}_{\leq 1}$, equipped with a GHM/history differential Q_{GHM} , and let $\widetilde{\mathcal{E}}$ extend $\mathcal{E}$ to the domain needed below.

Open obligation

AVC-2 (Derived Hardy Completion / No-Loss). Construct $\widetilde{\mathcal{S}}_{\leq 1}$, Q_{GHM} , and $\widetilde{\mathcal{E}}$ zero-blindly such that:

$$\nabla_{\text{ar}}(\mathcal{S}_{\leq 1} \cap \text{Dom } \nabla_{\text{ar}}) \subseteq \mathcal{S}_{\leq 1} + \text{im } Q_{\text{GHM}}, \quad (62)$$

$$\widetilde{\mathcal{E}}(\mathcal{S}_{\leq 1} + \text{im } Q_{\text{GHM}})^{H_{1/2}^2} = \overline{\mathcal{E}(\mathcal{S}_{\leq 1})}^{H_{1/2}^2} = \Delta, \quad (63)$$

and $\mathcal{E}(\mathcal{S}_{\leq 1} \cap \text{Dom } \nabla_{\text{ar}})$ is a graph core for ∂_s restricted to Δ .

Equation (62) is the *derived source invariance* statement. Equation (63) is the *no-new-normal-direction* statement: exact/history completion may enlarge the source complex, but it must not enlarge the closed Hardy image beyond Δ .

7.3 Equivalent normal-quotient formulation

Let

$$N_\Delta := H_{1/2}^2 / \Delta \quad (64)$$

be the normal quotient. AVC-2 can be read as saying that the exact correction has zero normal class and the core connection is tangent to Δ after completion:

$$[\tilde{\mathcal{E}}(Q_{\text{GHM}}\eta)]_{N_\Delta} = 0, \quad (65)$$

and, for the source core,

$$[\partial_s \mathcal{E}\varphi]_{N_\Delta} = [\mathcal{E}\nabla_{\text{ar}}\varphi]_{N_\Delta} = 0. \quad (66)$$

This is precisely the point where a purely algebraic Q -exactness statement must survive an infinite dimensional analytic completion.

8 AVC-1 + AVC-2 imply Hardy derivative invariance

Theorem / Proposition

Proposition (Derived closure $\Rightarrow$ derivative invariance). Assume AVC-1 and AVC-2. Then

$$\partial_s(\Delta \cap \text{Dom } \partial_s) \subseteq \Delta. \quad (67)$$

Proof. Take $F \in \Delta \cap \text{Dom } \partial_s$. By the graph-core clause in AVC-2 there exists a sequence $\varphi_n \in \mathcal{S}_{\leq 1} \cap \text{Dom } \nabla_{\text{ar}}$ such that

$$\mathcal{E}\varphi_n \rightarrow F, \quad \partial_s \mathcal{E}\varphi_n \rightarrow \partial_s F \quad (68)$$

in $H_{1/2}^2$. By AVC-1,

$$\partial_s \mathcal{E}\varphi_n = \tilde{\mathcal{E}}\nabla_{\text{ar}}\varphi_n. \quad (69)$$

By (62), each $\nabla_{\text{ar}}\varphi_n$ belongs to $\mathcal{S}_{\leq 1} + \text{im } Q_{\text{GHM}}$. By (63), its image belongs to the closed subspace Δ . Since Δ is closed and the sequence converges to $\partial_s F$, one has $\partial_s F \in \Delta$. $\square$

9 Derivative invariance kills the bad inner factor

Let $\Delta = BH_{1/2}^2$ as in (30). For g in a suitable analytic domain,

$$\partial_s(Bg) = Bg' + B'g. \quad (70)$$

The first term belongs to $BH_{1/2}^2$. The normal component is therefore represented by $B'g$.

Suppose B has a zero ρ of multiplicity $m \geq 1$:

$$B(s) = (s - \rho)^m u(s), \quad u(\rho) \neq 0. \quad (71)$$

Choose g with $g(\rho) \neq 0$. Then

$$B'(s)g(s) = m(s - \rho)^{m-1}u(\rho)g(\rho) + O((s - \rho)^m), \quad (72)$$

so $B'g$ vanishes to order exactly $m - 1$. But every element of $BH_{1/2}^2$ vanishes at ρ to order at least m . Hence

$$B'g \notin BH_{1/2}^2. \quad (73)$$

Therefore:

Theorem / Proposition**Derivative-invariance lemma.** If

$$\partial_s(\Delta \cap \text{Dom } \partial_s) \subseteq \Delta, \quad (74)$$

then B has no zero in $\Re s > 1/2$. Under the Burnol factorization input used in the Naive Proof, the remaining singular/exponential inner factors are absent/trivial, so normalization gives

$$\boxed{B = 1.} \quad (75)$$

Together with (31), this is the analytic endpoint of the soft-proof chain.

10 Soft Proof v0.97 theorem chain**Theorem / Proposition**

Soft Proof v0.97 (conditional only on AVC-2). Assume the frozen NCHG/ray identities, the Burnol Hardy factorization endpoint, and AVC-2. Then the Riemann Hypothesis follows.

Proof chain. AVC-1 is already proved explicitly:

$$\partial_s \mathcal{E} = \mathcal{E} \nabla_{\text{ar}}. \quad (76)$$

AVC-2 promotes this core identity to the closed Hardy subspace:

$$\partial_s \Delta \subseteq \Delta. \quad (77)$$

The derivative-invariance lemma gives

$$B \text{ has no zeros.} \quad (78)$$

The Burnol inner-factor audit gives

$$B = 1. \quad (79)$$

Finally the Burnol closure criterion gives

$$\boxed{\text{RH.}} \quad (80)$$

In one line:

$$\boxed{\text{AVC-1} + \text{AVC-2} \implies \partial_s \Delta \subseteq \Delta \implies B = 1 \implies \text{RH.}} \quad (81)$$

Why this is v0.97 rather than v1.0. The differential identity that used to be postulated as GRV has been derived. The remaining premise AVC-2 is no longer a vague physical-sector condition: it is an explicit, zero-blind graph-closure and no-loss theorem. It is nevertheless still unproved, and it is strong enough to carry the remaining global proof burden. Therefore the chain is not yet an unconditional proof.

11 Minimal one-vector shadow

The Naive Proof observed that full derivative invariance is stronger than the final multiplicity contradiction actually needs. Let

$$g_0(s) = \frac{1}{(s+1)^2}. \quad (82)$$

If $Bg_0 \in \text{Dom } \partial_s$ and

$$(Bg_0)' \in BH_{1/2}^2, \quad (83)$$

then the same multiplicity argument forces B to be zero-free, hence $B = 1$, hence RH.

Thus AVC-2 has a very small analytic shadow: it must imply (83). This is useful for falsification and numerical/domain tests, but it must not be promoted to an independent “proof” unless derived zero-blindly from the source/history completion.

12 Non-circularity audit

12.1 What is zero-blind

The following objects do not use nontrivial zero locations in their definitions:

- the seed algebra A and its history complex;
- q_r, h_r, b_r, H, J and the transverse exactness relations;
- the Mellin transform (33);
- $V(s) = (s - 1)/s$;
- L, R , and P_Δ ;
- ∇_{ar} and C_{tr} ;
- the source-to-Hardy map $\mathcal{E}$ and the closed subspace Δ as an image/closure object;
- the desired derived completion in AVC-2, provided it is constructed from history/arithmetic data rather than from the zero set.

12.2 Why AVC-2 is still strong

AVC-2 implies derivative invariance of $BH_{1/2}^2$, and derivative invariance kills every nonconstant Blaschke factor by the elementary multiplicity argument. Therefore AVC-2 is not a small technicality. Its value is that it is now an explicit mathematical statement with independently inspectable subclaims rather than a restatement “all zeros are critical”.

12.3 Forbidden shortcuts

Falsification guard

The following do *not* count as proofs of AVC-2:

1. defining $Q_{\text{GHM}}, \widetilde{S_{\leq 1}},$ or $\widetilde{\mathcal{E}}$ using nontrivial zero ordinates;
2. assuming $\partial_s \Delta \subseteq \Delta$ directly;
3. assuming $(Bg_0)' \in BH_{1/2}^2$ directly and relabelling it “physicality”;
4. identifying formal Q -exactness on $C_r \widehat{\otimes} \mathcal{D}_0$ with Hardy-closure exactness without a graph/domain theorem;
5. using analytic continuation of $V\zeta$ as if it automatically preserved the same operator domain and range closure;
6. replacing the global Burnol source/closure problem by an independent direct sum of local prime

blocks;

7. claiming that full NCHG BV anomaly cancellation is already proved.

13 False-negative guard

The following failures would narrow the route but would not automatically refute the whole program:

- The particular core $C_c^\infty(0, \infty)$ may be too small; AVC-1 can be moved to another common Mellin core if the operator identities survive.
- Raw ∇_{ar} need not preserve $S_{\leq 1}$ strictly; AVC-2 only asks for preservation modulo the exact/history sector.
- Strict equality may be replaced by a chain-homotopy/derived equality if the induced normal class and any determinant torsion are controlled.
- A naive positive L^2 quotient may erase generalized zero modes; this does not by itself kill a Hardy, Frechet, distributional, or determinant-line completion.
- Failure of one proposed Q_{GHM} does not exclude another zero-blind derived source completion with the same low-arity transverse prototype.

14 Dependency graph at v0.97

NCHG history/ray exactness	$\implies$	$\mathbf{C}_{\text{tr}}$ has an explicit Q -exact lift
Prime + pole data	$\implies$	$\nabla_{\text{ar}} = \mathbf{L} + \mathbf{R} - \mathbf{P}_\Lambda$
Normalized arithmetic map $\mathcal{E}$	$\xRightarrow{\text{AVC-1}}$	$\partial_s \mathcal{E} = \mathcal{E} \nabla_{\text{ar}}$
$\partial_s \mathcal{E} = \mathcal{E} \nabla_{\text{ar}}$	$\xRightarrow{\text{AVC-2}}$	$\partial_s \Delta \subseteq \Delta$
$\partial_s \Delta \subseteq \Delta$	$\implies$	$B = 1 \implies \text{RH}$

The single dashed/open passage is the analytic completion of the exact arithmetic/history correction into the closed Hardy image.

15 Proof certificate

ID	Claim	Status	Evidence / exact formula
SP97-01	History differential	SOURCE-EXACT	$d^2 = 0, d\alpha = [\ , \]$.
SP97-02	Ray contraction	VERIFIED	$q_r^2 = 0, q_r h_r + h_r q_r = I$.
SP97-03	Transverse scalar exactness	VERIFIED	$\{Q_\delta, h_r\} = \delta I$.
SP97-04	Cartan exactness	VERIFIED	$\{Q_\delta, b_r\} _V = \delta H$.
SP97-05	History Cartan boundary	VERIFIED	$d_{\text{hist}} \mathbf{b}_H = H$.
SP97-06	Burnol Hardy factorization	EXTERNAL INPUT	$\Delta = BH_{1/2}^2$, carried from Naive Proof.

ID	Claim	Status	Evidence / exact formula
SP97-07	Burnol endpoint	EXTERNAL INPUT	$B = 1 \Leftrightarrow \text{RH in the chosen causality/closure formulation.}$
SP97-08	Mellin transform of $\mathbf{L}$	PROVED HERE	$\widehat{\mathbf{L}\varphi} = \partial_s \widehat{\varphi}.$
SP97-09	Mellin transform of $\mathbf{R}$	PROVED HERE	$\widehat{\mathbf{R}\varphi} = (1/(s-1) - 1/s)\widehat{\varphi} = (V'/V)\widehat{\varphi}.$
SP97-10	Mellin transform of $\mathbf{P}_\Lambda$	PROVED HERE	$\widehat{\mathbf{P}_\Lambda\varphi} = -(\zeta'/\zeta)\widehat{\varphi}.$
SP97-11	AVC-1	PROVED HERE	$\partial_s \mathcal{E} = \mathcal{E} \nabla_{\text{ar}}$ on the zero-blind Mellin core.
SP97-12	Transverse arithmetic operator	EXPLICIT	$\mathbf{C}_{\text{tr}} = \mathbf{R} - \mathbf{P}_\Lambda.$
SP97-13	Formal exact lift of $\mathbf{C}_{\text{tr}}$	VERIFIED ON CORE	$I \otimes \mathbf{C}_{\text{tr}} = \{q_r \otimes \mathbf{C}_{\text{tr}}, h_r \otimes I\}.$
SP97-14	AVC-2	OPEN BOSS	Derived source invariance + no-loss closure + graph-core compatibility, Eqs. (62)-(63).
SP97-15	AVC-2 $\Rightarrow$ derivative invariance	PROVED HERE	Graph-core approximation plus AVC-1.
SP97-16	Derivative invariance $\Rightarrow B$ zero-free	PROVED HERE	Local multiplicity drop from m to $m-1$ in $B'g$.
SP97-17	Soft-proof chain	CONDITIONAL	AVC-1 + AVC-2 $\Rightarrow B = 1 \Rightarrow \text{RH}.$

16 What would qualify as Soft Proof v1.0?

The upgrade criterion is deliberately strict. Version 1.0 requires a zero-blind proof of AVC-2, including:

1. an explicit derived source space $\widetilde{\mathcal{S}_{\leq 1}}$ and differential Q_{GHM} ;
2. a domain theorem for ∇_{ar} on the Burnol source core;
3. a proof of derived invariance (62);
4. a proof of no-loss closure (63);
5. a graph-core theorem allowing ∂_s to pass from the source core to the closed Hardy subspace;
6. an audit that no hidden zero data enters the construction;
7. a check that any chain-homotopy/torsion correction does not alter the normal quotient relevant to the derivative-invariance lemma.

Once these are proved, the remaining RH implication is the already-closed chain (81). Until then, v0.97 must remain explicitly conditional.

A Matrix verification of the ray identities

From (13) and (14), direct multiplication gives

$$q_r h_r = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad h_r q_r = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad (84)$$

so their sum is I_4 . Likewise

$$q_r b_r + b_r q_r = \text{diag}(1, 1, -1, -1). \quad (85)$$

The middle 2×2 block is exactly $H = \text{diag}(1, -1)$.

B Detailed Mellin calculation for the prime-power operator

For $\varphi \in C_c^\infty(0, \infty)$,

$$\mathcal{M}(\mathbf{P}_\Lambda \varphi)(s) = \int_0^\infty \left(\sum_{n \geq 2} \Lambda(n) \varphi(nx) \right) x^s \frac{dx}{x} \quad (86)$$

$$= \sum_{n \geq 2} \Lambda(n) \int_0^\infty \varphi(nx) x^s \frac{dx}{x} \quad (87)$$

$$= \sum_{n \geq 2} \Lambda(n) n^{-s} \int_0^\infty \varphi(y) y^s \frac{dy}{y} \quad (88)$$

$$= \left(\sum_{n \geq 2} \frac{\Lambda(n)}{n^s} \right) \widehat{\varphi}(s) \quad (89)$$

$$= -\frac{\zeta'(s)}{\zeta(s)} \widehat{\varphi}(s), \quad \Re s > 1. \quad (90)$$

The pointwise finiteness of the source sum and compact support justify the exchange of sum and integral on the core. The meromorphic continuation of the multiplier is not, by itself, a domain theorem for the Hardy completion; that distinction is exactly why AVC-2 remains open.

C Detailed Mellin calculation for the pole correction

Write $\mathbf{R} = \mathbf{R}_1 - \mathbf{R}_2$ with

$$\mathbf{R}_1 \varphi(x) = \frac{1}{x} \int_x^\infty \varphi(y) dy, \quad \mathbf{R}_2 \varphi(x) = \int_x^\infty \varphi(y) \frac{dy}{y}. \quad (91)$$

For $\Re s > 1$,

$$\mathcal{M}(\mathbf{R}_1 \varphi)(s) = \int_0^\infty \int_x^\infty \varphi(y) x^{s-2} dy dx \quad (92)$$

$$= \int_0^\infty \varphi(y) \int_0^y x^{s-2} dx dy \quad (93)$$

$$= \frac{1}{s-1} \widehat{\varphi}(s), \quad (94)$$

and

$$\mathcal{M}(\mathbf{R}_2\varphi)(s) = \int_0^\infty \int_x^\infty \varphi(y)y^{-1}x^{s-1} dy dx \quad (95)$$

$$= \int_0^\infty \frac{\varphi(y)}{y} \int_0^y x^{s-1} dx dy \quad (96)$$

$$= \frac{1}{s} \widehat{\varphi}(s). \quad (97)$$

Thus

$$\mathcal{M}(\mathbf{R}\varphi)(s) = \frac{1}{s(s-1)} \widehat{\varphi}(s) = \frac{V'(s)}{V(s)} \widehat{\varphi}(s) \quad (98)$$

for $V(s) = (s-1)/s$.

D Provenance and references

Project-internal sources

- P1. *RH-GHM-NCHG Research Ledger*, 12 August 2026. Core algebra, history complex, time-ray Hodge theorem, arithmetic globalization, falsification ledger.
- P2. *RH-GHM-NCHG Supplemental I: C1 Update, C1 Faithfulness and C2 Global Factorization Specification*, 12 August 2026. C1-E/I/F hierarchy and C2 contract.
- P3. *RH-GHM-NCHG C2 Ledger*, 12 August 2026. Explicit transverse exactness, history/ray square, local superconnection, determinant-line correction.
- P4. *RH-GHM-NCHG Naive Conditional Proof: Global Relative Verticality Version*, 12 August 2026. Burnol Hardy endpoint, GRV conditional theorem, multiplicity lemma.
- P5. *NCHG v0.2: Nonassociative Causal-Holonomy Gravity*, 2026. Associahedral history resolution, BRST/BV backbone, chronology structures.

External analytic sources carried through the project documents

- E1. J.-F. Burnol, *An adelic causality problem related to abelian L-functions*, arXiv:math/0001013 (2000). Used in the Naive Proof for the Hardy-space inner-factor/causality endpoint.
- E2. J.-F. Burnol, *The explicit formula and the conductor operator*, arXiv:math/9901051 (1999). Structural precedent for local conductor/explicit-formula analysis.
- E3. R. Meyer, *A spectral interpretation for the zeros of the Riemann zeta function*, arXiv:math/0412277 (revised 2013). Structural precedent for zero-blind arithmetic scaling/character realizations.

Final freeze. Soft Proof v0.97 does not assert that AVC-2 is proved. The main advance over the Naive Proof is that the former GRV black box has been decomposed into an explicit proved arithmetic connection identity (AVC-1) and a single sharply formulated derived Hardy-completion theorem (AVC-2). The latter is the only new global premise in the v0.97 RH chain.